# MATH50003 Numerical Analysis

## Revision

Dr Sheehan Olver

# Revision Sheet

6 Approximate Fourier coefficients

1 Divided Differences

4 Cholesky factorisation

5 2x2 QR

7 Chebyshev

**Problem 6** For the function $f(\theta) = \sin 3\theta$, state explicit formulae for its Fourier coefficients

$$\hat{f}_k := \frac{1}{2\pi} \int_0^{2\pi} f(\theta) \mathrm{e}^{-\mathrm{i}k\theta} \mathrm{d}\theta$$

and their discrete approximation:

$$\hat{f}_k^n := \frac{1}{n} \sum_{j=0}^{n-1} f(\theta_j) \mathrm{e}^{-\mathrm{i}k\theta_j}.$$

for *all* integers $k$, $n = 1, 2, \ldots$, where $\theta_j = 2\pi j/n$.

**Problem 1** Show that

$$\frac{f(x+2h)-f(x-h)}{3h} = f'(x) + \delta$$

where

$$|\delta| \leq 5Mh/6$$

for $M = \sup_{x-h \leq t \leq x+2h} |f''(t)|$

**Problem 4** Use the Cholesky factorisation to determine whether the following matrix is symmetric positive definite:

$$\begin{bmatrix} 2 & 2 & 1 \\ 2 & 3 & 2 \\ 1 & 2 & 2 \end{bmatrix}$$

**Problem 5** Compute the QR factorisation of

$$\begin{bmatrix} -3 & 1 \\ 4 & 2 \end{bmatrix}$$

**Problem 7(a)** Prove that the second kind Chebyshev polynomials

$$U_n(x) := \frac{\sin(n+1)\mathrm{acos}\, x}{\sqrt{1-x^2}}$$

satisfy for any polynomial $p(x)$ of degree $m < n$

$$\int_{-1}^{1} U_n(x)p(x)\sqrt{1-x^2}\mathrm{d}x = 0.$$

**Problem 7(b)** Use this to prove that

$$C_n(x) := U'_{n+1}(x)$$

are orthogonal with respect to the inner product

$$\int_{-1}^{1} f(x)g(x)(1-x^2)^{3/2}\mathrm{d}x.$$